

PE-EC801A — Antennas and Propagation

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Module 2 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Infinitesimal Dipole

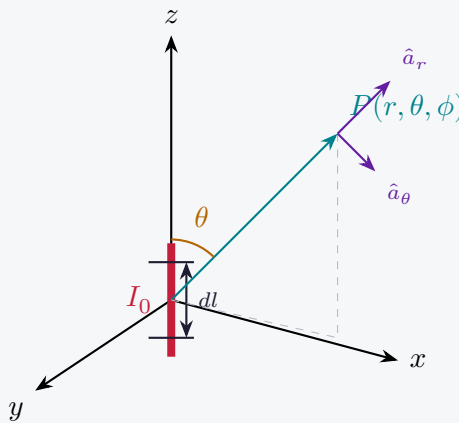
Definition of Infinitesimal Dipole

An **infinitesimal dipole** (also known as a Hertzian dipole) is a thin linear antenna whose length dl is extremely small compared to the wavelength ($dl \ll \lambda$) and whose current is assumed to be uniform ($I(z) = I_0 = \text{constant}$) over its entire length.

1.1 Coordinate System and Potential Derivation

Consider a wire of length dl placed symmetrically at the origin along the z -axis. Since the current is along z , the current density is $\vec{J} = \hat{a}_z I_0 \delta(x') \delta(y')$ for $-dl/2 \leq z' \leq dl/2$.

Infinitesimal Dipole Geometry



The magnetic vector potential $\vec{A}$ at a distance r (since $r \gg dl$) is:

$$\vec{A}(r) = \hat{a}_z \frac{\mu I_0 dl}{4\pi r} e^{-jkr}$$

Using coordinate transformations, the spherical components of $\vec{A}$ are:

$$\begin{aligned} A_r &= A_z \cos \theta = \frac{\mu I_0 dl \cos \theta}{4\pi r} e^{-jkr} \\ A_\theta &= -A_z \sin \theta = -\frac{\mu I_0 dl \sin \theta}{4\pi r} e^{-jkr} \\ A_\phi &= 0 \end{aligned}$$

1.2 Electromagnetic Field Equations

By applying Maxwell's relations ($\vec{H} = \frac{1}{\mu} \nabla \times \vec{A}$ and $\vec{E} = \frac{1}{j\omega\epsilon} \nabla \times \vec{H}$), we obtain the complete field equations:

$$\begin{aligned} H_\phi &= j \frac{k I_0 dl \sin \theta}{4\pi r} \left[1 + \frac{1}{jkr} \right] e^{-jkr} \\ E_r &= \frac{2 I_0 dl \eta \cos \theta}{4\pi r^2} \left[1 + \frac{1}{jkr} \right] e^{-jkr} \end{aligned}$$

$$E_\theta = j \frac{k I_0 dl \eta \sin \theta}{4\pi r} \left[1 + \frac{1}{jkr} - \frac{1}{(kr)^2} \right] e^{-jkr}$$

where $\eta = \sqrt{\mu/\epsilon} \approx 120\pi \approx 377 \Omega$ is the intrinsic impedance of free space. Other components $H_r = H_\theta = E_\phi = 0$.

1.3 Far-Field Approximation and Radiation Resistance

For the far field ($kr \gg 1$), terms containing $1/(kr)^2$ and $1/(kr)^3$ can be neglected. The fields simplify to:

$$H_\phi \approx j \frac{I_0 k dl \sin \theta}{4\pi r} e^{-jkr}$$

$$E_\theta \approx j \eta \frac{I_0 k dl \sin \theta}{4\pi r} e^{-jkr}$$

The total power radiated is found by integrating the Poynting vector over a closed sphere:

$$P_{rad} = \int_0^{2\pi} \int_0^\pi \frac{1}{2} \text{Re}(E_\theta H_\phi^*) r^2 \sin \theta d\theta d\phi = 40\pi^2 I_0^2 \left(\frac{dl}{\lambda} \right)^2$$

Since $P_{rad} = \frac{1}{2} I_0^2 R_r$, the radiation resistance R_r of an infinitesimal dipole is:

$$R_r = 80\pi^2 \left(\frac{dl}{\lambda} \right)^2$$

Note: The Short Dipole

For a **short dipole** (length $l \ll \lambda$, but current decreases linearly from I_0 at the center to 0 at the ends), the average current is $I_0/2$. The radiation resistance is 1/4 of the infinitesimal dipole:

$$R_r = 20\pi^2 \left(\frac{l}{\lambda} \right)^2$$

2. Finite-Length Dipole

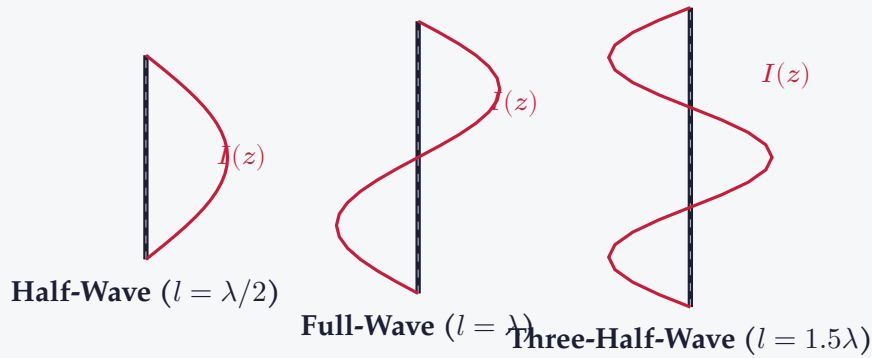
When the antenna length is comparable to the wavelength, the current is no longer uniform. It forms a standing wave with a sinusoidal distribution (assuming thin wire and zero current at the ends).

2.1 Current Distribution

For a dipole of length l placed along the z -axis, the current distribution is:

$$I(z') = I_0 \sin \left[k \left(\frac{l}{2} - |z'| \right) \right], \quad -l/2 \leq z' \leq l/2$$

Current Distributions on Dipoles



2.2 Far-Field Radiation Pattern

By treating the finite dipole as an infinite sum of infinitesimal dipoles and integrating the contributions, the far-field electric field is:

$$E_{\theta} \approx j\eta \frac{I_0 e^{-jkr}}{2\pi r} \left[\frac{\cos\left(\frac{kl}{2} \cos \theta\right) - \cos\left(\frac{kl}{2}\right)}{\sin \theta} \right]$$

$$H_{\phi} = \frac{E_{\theta}}{\eta}$$

2.3 The Half-Wave Dipole ($l = \lambda/2$)

Setting $l = \lambda/2$ (and thus $kl/2 = \pi/2$) in the field equations:

$$E_{\theta} \approx j\eta \frac{I_0 e^{-jkr}}{2\pi r} \left[\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right]$$

- **Radiation Resistance:** $R_r \approx 73 \Omega$.
- **Input Impedance:** $Z_{in} \approx 73 + j42.5 \Omega$ (inductive reactant due to end effects).
- **Resonant Length:** To make the input impedance purely real ($Z_{in} \approx 73 + j0 \Omega$), the dipole is shortened slightly by about 5% ($l \approx 0.47\lambda - 0.48\lambda$).
- **Directivity (D_0):** 1.64 (2.15 dBi).
- **HPBW:** 78° .

3. Linear Elements Near Conductors (Image Theory)

To analyze antennas placed near flat, perfectly conducting ground planes, we replace the ground plane with an equivalent virtual antenna below the plane. This is called **Image Theory**.

Image Theory Equivalents



3.1 Boundary Rules for Image Currents

- **Vertical Element:** A vertical current element at distance h above a ground plane has an image element of the same length and direction (in-phase) at distance h below the plane.
- **Horizontal Element:** A horizontal current element at distance h above a ground plane has an image element of the same length but opposite direction (out-of-phase) at distance h below the plane.

4. Dipoles for Mobile Communication (Monopoles)

For mobile handsets and base stations, it is often impractical to use a symmetric double-sided dipole. Instead, a **monopole antenna** (a single wire placed perpendicular to a conducting ground plane) is used.

4.1 The Quarter-Wave Monopole ($l = \lambda/4$)

Using image theory, a monopole of length $l = \lambda/4$ on a perfect ground plane is equivalent to a half-wave dipole ($l = \lambda/2$) in free space, but only for the upper half-space ($z \geq 0$).

- **Fields ($z \geq 0$):** Identical to half-wave dipole fields.
- **Fields ($z < 0$):** Zero (shielded by ground plane).
- **Radiated Power ($P_{rad,mono}$):** Since it radiates only into the upper hemisphere, the radiated power for the same input current is half that of a dipole:

$$P_{rad,monopole} = \frac{1}{2} P_{rad,dipole}$$

- **Radiation Resistance ($R_{r,mono}$):**

$$R_{r,monopole} = \frac{1}{2} R_{r,dipole} = \frac{73}{2} \approx 36.5 \Omega$$

- **Directivity (D_{mono}):** Since the power is focused into half the volume, the directivity is double that of a dipole:

$$D_{monopole} = 2 \times D_{dipole} = 2 \times 1.64 = 3.28 \quad (5.16 \text{ dBi})$$

5. Small Circular Loop Antenna

Definition of Small Circular Loop

A **small circular loop antenna** is a loop antenna of radius a whose circumference is much smaller than the wavelength ($2\pi a \ll \lambda$), resulting in a uniform current distribution $I(z) = I_0$ along the loop.

5.1 Far-Field Radiated Fields

By using the magnetic vector potential $\vec{A} = \hat{a}_\phi A_\phi$, the far fields for a loop with area $S = \pi a^2$ are:

$$E_\phi \approx \frac{\eta k^2 I_0 S \sin \theta}{4\pi r} e^{-jkr}$$

$$H_\theta \approx -\frac{E_\phi}{\eta} \approx -\frac{k^2 I_0 S \sin \theta}{4\pi r} e^{-jkr}$$

5.2 Radiation Resistance

The power radiated by the loop is integrated, leading to the radiation resistance:

$$R_r = 31200 \left(\frac{S}{\lambda^2} \right)^2 = 31200 \left(\frac{\pi a^2}{\lambda^2} \right)^2 \approx 320\pi^4 \left(\frac{a}{\lambda} \right)^4$$

Important Note: Small Loop as a Magnetic Dipole

The radiation pattern of a small loop has a torus/donut shape ($f(\theta) = \sin \theta$), which is identical to the pattern of an infinitesimal electric dipole. Thus, a small loop behaves as a **magnetic dipole** perpendicular to the plane of the loop. However, its radiation resistance is much smaller than that of an electric dipole of similar size, making it a less efficient radiator (often used primarily for receiving, e.g., AM radio).

Quick Revision — Module 2

Wire and Loop Antennas | PE-EC801A

Key Formulas and Metrics at a Glance

Infinitesimal Dipole R_r	$R_r = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2$ ($dl \ll \lambda$, uniform current)
Short Dipole R_r	$R_r = 20\pi^2 \left(\frac{l}{\lambda}\right)^2$ ($l \ll \lambda$, linear current)
Half-Wave Dipole R_r & D_0	$R_r \approx 73 \Omega$, $D_0 \approx 1.64$ (2.15 dBi)
Quarter-Wave Monopole R_r & D_0	$R_r \approx 36.5 \Omega$, $D_0 \approx 3.28$ (5.16 dBi)
Small Loop R_r	$R_r \approx 31200 \left(\frac{S}{\lambda^2}\right)^2$ ($S = \pi a^2$)
Dipole Current Profile	$I(z) = I_0 \sin[k(l/2 - z)]$
Resonant Dipole Length	$l_{resonant} \approx 0.47\lambda - 0.48\lambda$ (removes $+j42.5 \Omega$ reactance)

One-Line Definitions

- **Hertzian Dipole:** A theoretical, infinitely thin dipole antenna of length much smaller than a wavelength, carrying uniform current.
- **Resonant Antenna:** An antenna whose input impedance is purely resistive ($X_{in} = 0$) at its operating frequency.
- **Image Theory:** A mathematical method replacing a conducting boundary with virtual mirror-image current sources.
- **Quarter-Wave Monopole:** A single vertical wire of length $\lambda/4$ placed above a conducting ground plane.
- **Magnetic Dipole:** The dual of an electric dipole; represented physically by a small circular loop antenna.

Frequently Asked / Viva Questions

Q1. What is the difference between an infinitesimal dipole and a short dipole?

An infinitesimal dipole has a constant, uniform current along its length ($dl \ll \lambda$). A short dipole has current that drops off linearly from the center to zero at the ends ($l \ll \lambda$). Because of this current profile, the radiation resistance of a short dipole is one-quarter of that of an infinitesimal dipole.

Q2. Why does a half-wave dipole require shortening for resonance?

A physical half-wave dipole ($l = 0.5\lambda$) has an input impedance of $Z_{in} \approx 73 + j42.5 \Omega$. To eliminate the inductive reactance and make it resonant ($X_{in} = 0$), the antenna is shortened slightly to about $0.47\lambda - 0.48\lambda$.

Frequently Asked / Viva Questions (contd.)

Q3. Compare a half-wave dipole and a quarter-wave monopole antenna.

A quarter-wave monopole on a ground plane has half the radiation resistance ($36.5\ \Omega$ vs. $73\ \Omega$) and twice the directivity (3.28 vs. 1.64 , or $5.16\ \text{dBi}$ vs. $2.15\ \text{dBi}$) of a free-space half-wave dipole, because its radiation is restricted to the upper hemisphere.

Q4. Explain how the image of a horizontal dipole differs from a vertical dipole.

For a vertical dipole, the image current is in the same direction (in-phase). For a horizontal dipole, the image current is in the opposite direction (out-of-phase), which causes the radiated field to cancel at grazing angles (0° elevation).

Q5. Why are small loop antennas rarely used as transmitting antennas?

Small loop antennas have a very small radiation resistance (proportional to $(a/\lambda)^4$). This makes their radiation efficiency extremely low, as most input power is lost as heat in the conductor. They are, however, excellent for receiving applications (like AM radio) where noise levels are high and matching can be tuned.

Q6. Write the far-field expressions for a half-wave dipole.

$E_\theta = j120I_0 \frac{e^{-jkr}}{2r} \left[\frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} \right]$ and $H_\phi = \frac{E_\theta}{\eta}$. Other field components are zero.

Q7. What is the input impedance of a practical half-wave dipole at resonance?

At resonance, the input reactance is tuned to zero, and the input impedance is purely resistive: $Z_{in} \approx 73\ \Omega$.

Comparison of Wire and Loop Antennas

Parameter	Hertzian Dipole	Half-Wave Dipole	Small Loop
Length / Size	$dl \ll \lambda$	$l = \lambda/2$	Radius $a \ll \lambda/2\pi$
Current Profile	Uniform (I_0)	Sinusoidal ($I_0 \cos kz$)	Uniform (I_0)
Radiation Resistance	$80\pi^2(dl/\lambda)^2$	$73\ \Omega$	$31200(\pi a^2/\lambda^2)^2$
Directivity	1.5 ($1.76\ \text{dBi}$)	1.64 ($2.15\ \text{dBi}$)	1.5 ($1.76\ \text{dBi}$)
Field Type	Electric Dipole	Finite Electric Dipole	Magnetic Dipole